AlphaLens Platform

System Architecture, Agent Skills & Technical Operations Manual

This technical guide provides a comprehensive conceptual and operational breakdown of the **AlphaLens Autonomous Quantitative Research Terminal**. It details the multi-agent architecture, mathematical frameworks, inter-agent protocols, and UI skills.

1. Multi-Agent Orchestration Architecture

AlphaLens uses **LangGraph** to coordinate a stateful, directed multi-agent graph with iterative refinement loops and memory persistence.

Core Execution Flow (7 Agents / Nodes)

1. **Literature Agent:** Performs RAG semantic search over academic paper corpora (FAISS/ChromaDB) to generate structured hypotheses H = (predictor, target, direction, mechanism).
2. **Signal Generation Agent:** Computes 300+ technical/fundamental features, applies factor orthogonalization, and screens Information Coefficient (IC) & ICIR.
3. **Deep Learning Agent:** Trains multi-horizon forecasting models using Temporal Fusion Transformer (TFT), N-BEATS, and PatchTST ensemble.
4. **GNN Agent:** Leverages Graph Attention Networks (GAT) to capture cross-asset spatial

relationships and spillover dynamics.

5. **Causal Validation Agent:** Runs PC-Algorithm DAGs, Fast Causal Inference (FCI) PAGs (detecting latent confounders X ↔️ Y), 5-fold Double Machine Learning (DML), Partial R^2, and Rosenbaum bounds ($\Gamma = 2.0$).

6. **Backtesting Agent:** Out-of-sample simulation incorporating Kyle's λ market impact cost model $TC(q) = \sigma \sqrt{|q| / V\_ADV} \cdot sgn(q) \cdot P$ and survivorship bias delisting corrections.

7. **Portfolio Construction Agent:** Blends Black-Litterman market equilibrium views with model alphas, solving Mean-CVaR Optimization via CVXPY + CLARABEL.

2. Key Technical Concepts & Mathematical Frameworks

A. Signal Orthogonalization (§5.3)

To eliminate redundant risk factor exposure, candidate signals are projected onto the orthogonal complement of the existing factor matrix S:

```
s_orthogonal = s_i - S · (Sᵀ S)⁻¹ · Sᵀ s_i
```

B. Fast Causal Inference (FCI) & PAGs (§5.4)

Unlike standard DAG discovery which assumes no hidden variables, FCI relaxes causal sufficiency to output a **Partial Ancestral Graph (PAG)**. It detects unmeasured latent confounders (X ↔️ Y), preventing the agent from misidentifying common-cause correlations as direct causal alphas.

C. Partial R^2 Sensitivity Analysis (§5.4)

Quantifies the unique explanatory power of a signal against omitted variable bias:

```
Partial R² = (R²_full - R²_restricted) / (1 - R²_restricted)
```

Evaluates how strong an unobserved factor must be to nullify the estimated causal effect (Cinelli-Hazlett framework).

D. Mean-CVaR Optimization via CVXPY & CLARABEL (§10.1–10.2)

Solves Rockafellar-Uryasev linear programming formulation for portfolio weights w under concentration limits $w \leq w_{max}$:

```
min_{w, γ, z_s} γ + [1 / ((1 - α) S)] · Σ z_s
s.t. z_s ≥ -r_{p,s} - γ, z_s ≥ 0, μᵀ w ≥ μ_required, 1ᵀ w = 1
```

E. Inter-Agent Communication Protocol (§5.1)

Inter-agent messages are serialized using **Protocol Buffers (Protobuf)** following the message schema M = (sender, recipient, payload, timestamp, priority), persisted to PostgreSQL binary event logs for audit replayability.

3. Agent Memory & Prompt Engineering Framework (§11.2–11.3)

Memory Tier	Implementation	Purpose
Working Memory	LangGraph State Schema (TypedDict)	In-context transient execution state passed node-to-node.
Episodic Memory	PostgreSQL / SQLite Event Log	Historical decision trails and logged agent episodes.
Semantic Memory	ChromaDB / Vector Store	Associative academic facts and RAG literature embeddings.

Structured Prompt Template (§11.3)

All agent LLM calls follow a strict 4-part structure:

- `SYSTEM` Defines role, reasoning requirements, and target JSON output schema.
- `CONTEXT` Provides current hypothesis, extracted RAG passages, and past episodic decisions.
- `TASK` Specifies exact quantitative evaluation metrics and gating thresholds.
- `CONSTRAINTS` Enforces no look-ahead bias, mandatory statistical citations, and uncertainty flags.

4. User Interface & Data Ingestion Capabilities

- **Unified Choose File Uploader:** Accepts `.csv`, `.xlsx`, `.xls`, `.parquet` (market price data) and `.pdf`, `.txt` (research literature).
- **Dual Execution Modes:** Seamless toggle between *Chat & Explainer Mode* (direct Q&A) and *Quant Pipeline Execution Mode* (full 7-agent execution).
- **Visual Diagnostics:** Real-time rendering of IC/ICIR metrics, GAT cross-asset graphs, regime probabilities, and interactive Plotly asset allocation pies.